audio.dev
CONFERENCE
BRISTOL & ONLINE 2026

The Space Between Presets

Exploration of the Software Synth Presets Space

Shiva Sankar Kannan

Who I am



DREAMWORKS

TAU FILMS

2016 — 2019

Who I am



DREAMWORKS

TAU FILMS

Carnegie Mellon University
Entertainment Technology Center

2016 — 2019 — 2021

Who I am



DreamWorks

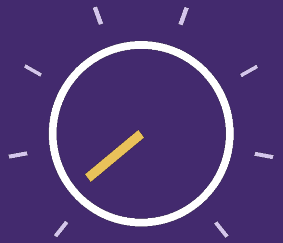
TAU FILMS

Carnegie Mellon University
Entertainment Technology Center

Dolby

2016 — 2019 — 2021 — now

Learning synths

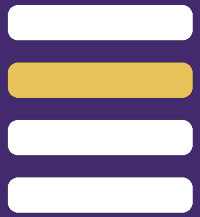


Knobs first

Learning synths



Knobs first

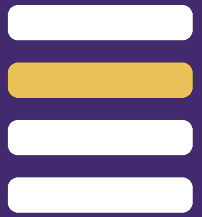


Presets first

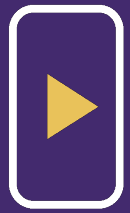
Learning synths



Knobs first



Presets first



Tutorials first

How I learn synths



Mid Key Squeal
Asymptote
Saw Difference Brass
LD Switch LFO Squelch 2
BASS Broken Circuit
Revokoto

How I learn synths



Mid Key Squeal

Asymptote

Saw Difference Brass

LD Switch LFO Squelch 2

BASS Broken Circuit

Revokoto

Preset libraries run to thousands of sounds



26,000+

Omnisphere 3



12,000+

V Collection Pro



1,200

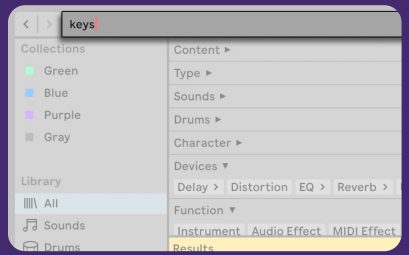
Diva



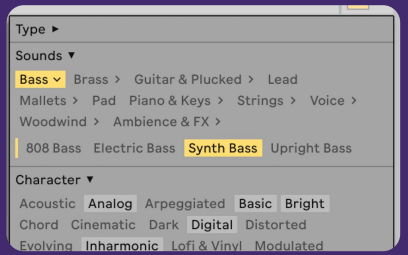
~600

Serum 2

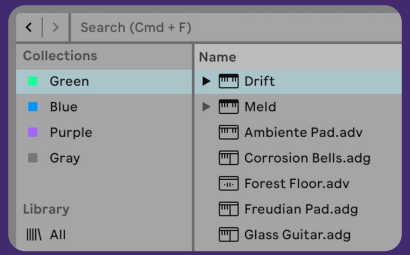
Preset search today



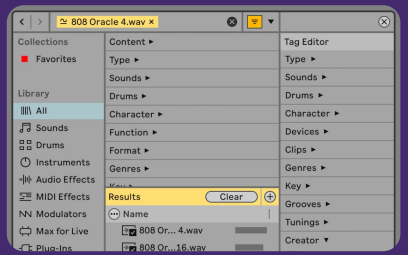
Search by name



Filter by tags



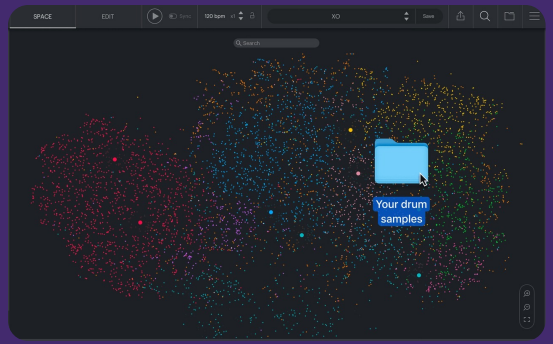
Favourites



Similar sounds

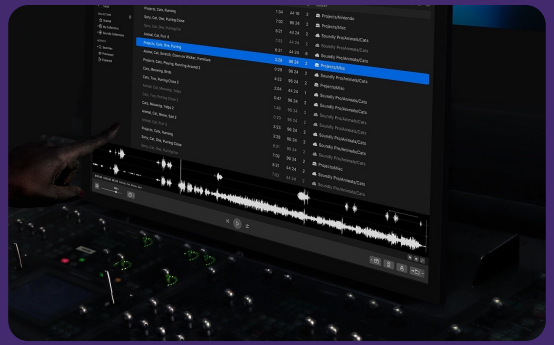
Screenshots: Ableton Live 12 manual

Techniques we could borrow



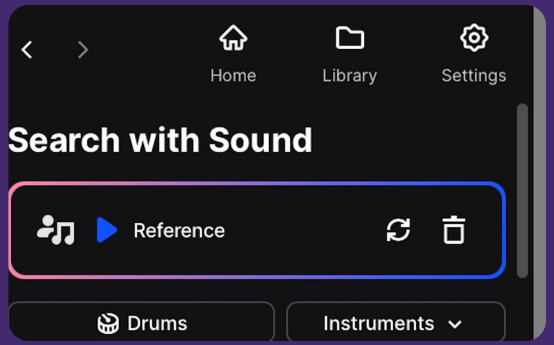
XLN XO

a map of similar sounds



Soundly

describe a sound



Splice

search with a sound

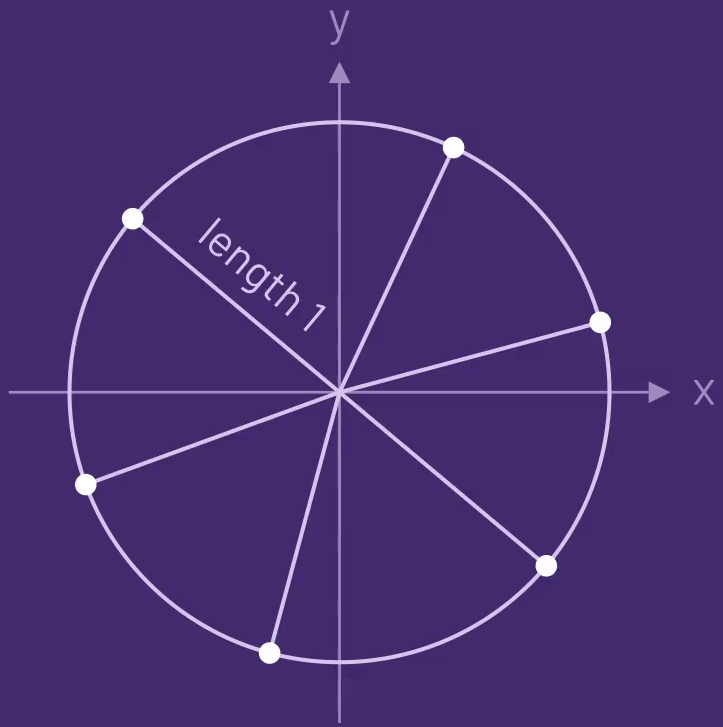
SynthScribe: Brade et al., IUI 2024

Images: xlnaudio.com, getsoundly.com, Ableton Live 12 manual

Cosine Similarity

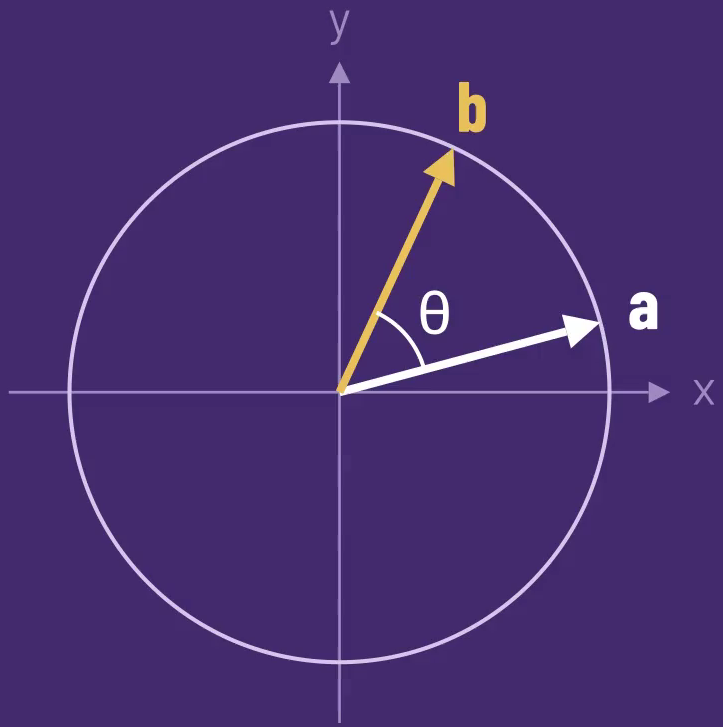
Animations: Manim Community Edition

Cosine Similarity



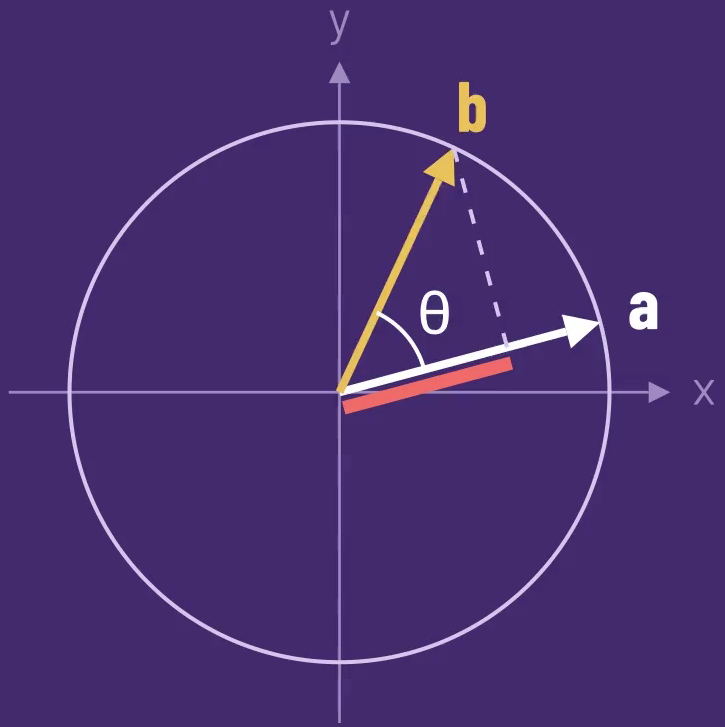
Animations: Manim Community Edition

Cosine Similarity



Animations: Manim Community Edition

Cosine Similarity

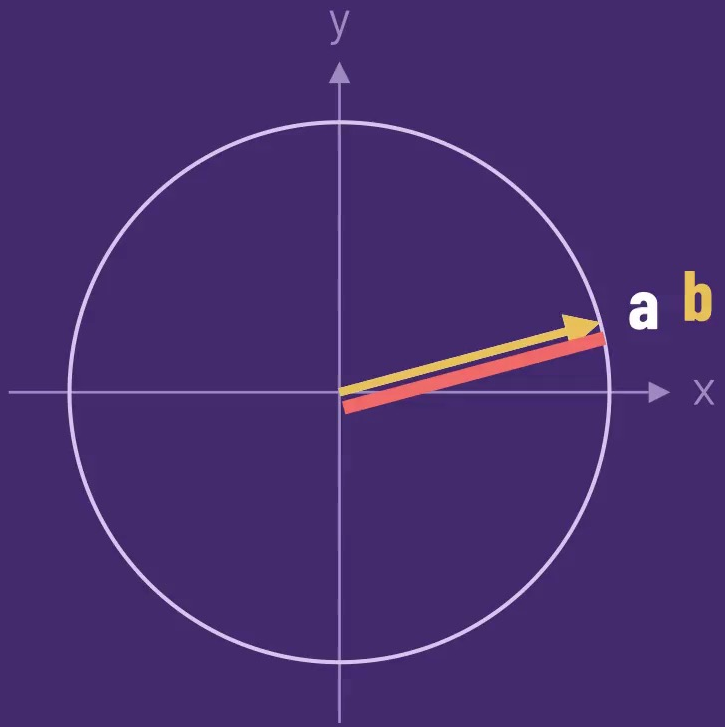


cos θ

0.64

Animations: Manim Community Edition

Cosine Similarity



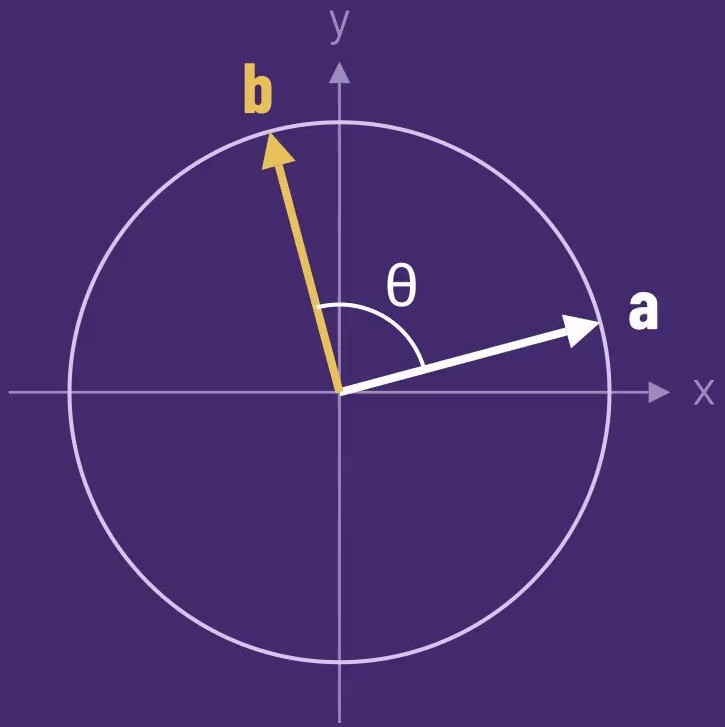
cos θ

1.00

same direction

Animations: Manim Community Edition

Cosine Similarity



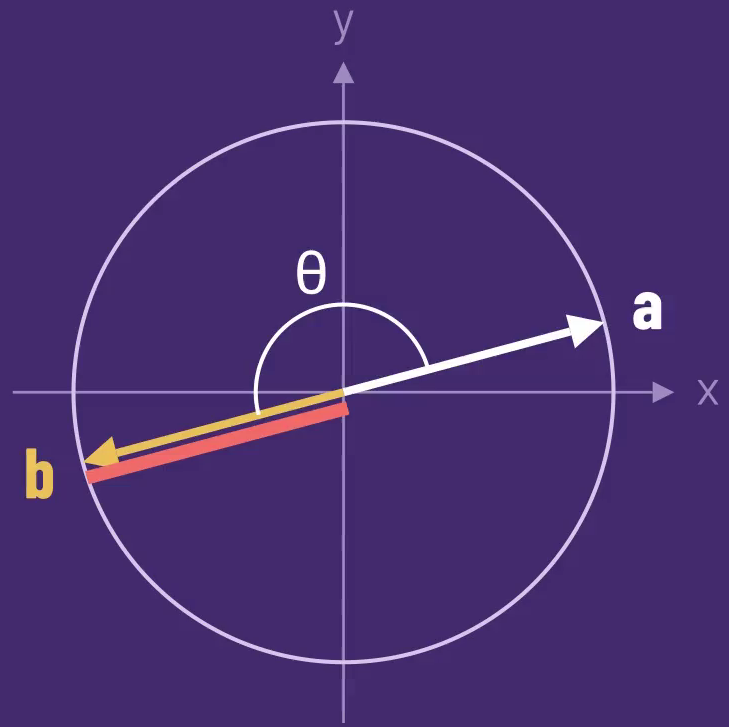
cos θ

0.00

90° apart

Animations: Manim Community Edition

Cosine Similarity



cos θ

−1.00

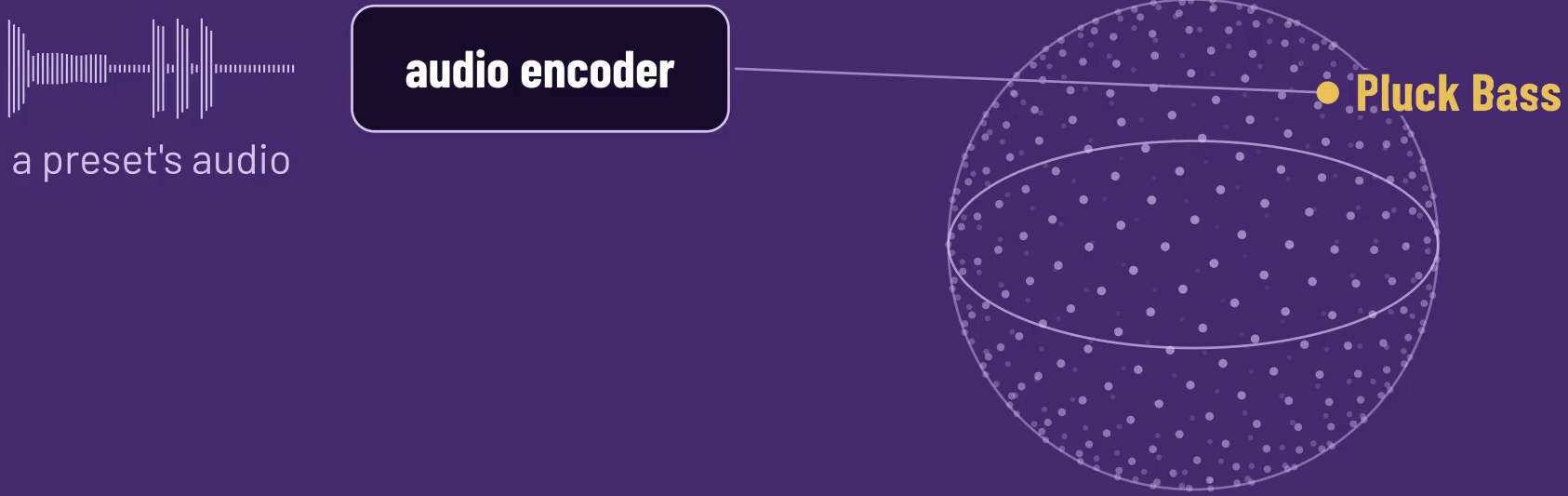
opposite

Animations: Manim Community Edition

CLAP puts text and audio in one space

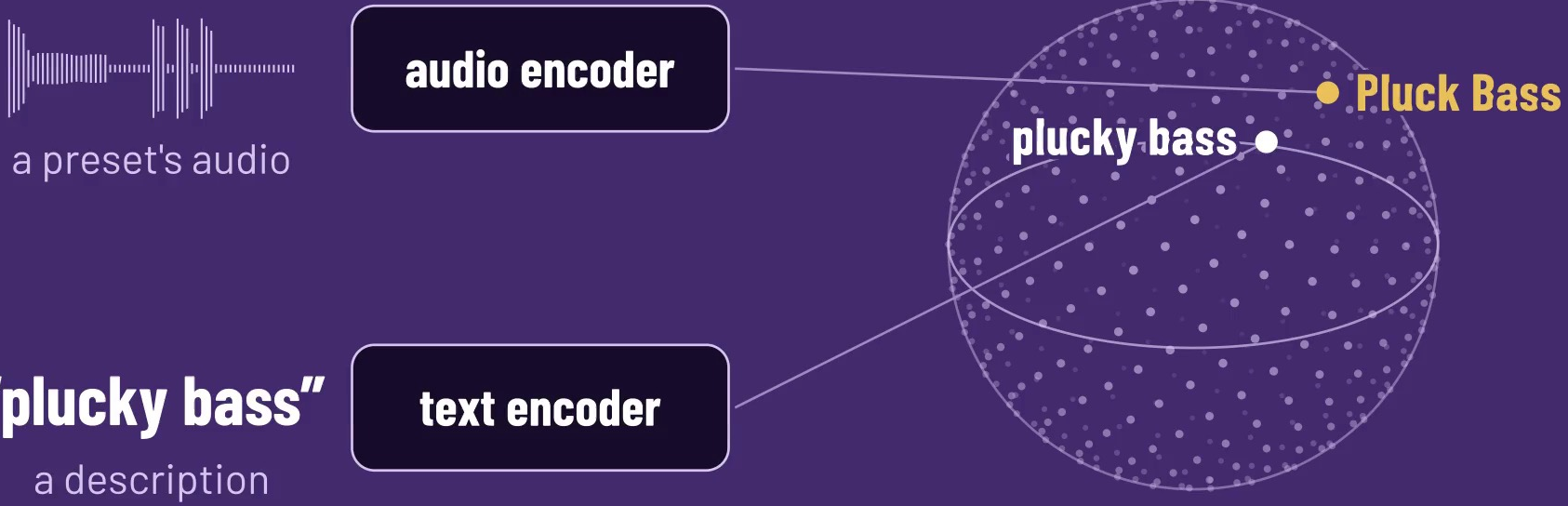
Animations: Manim Community Edition

CLAP puts text and audio in one space



Animations: Manim Community Edition

CLAP puts text and audio in one space



Animations: Manim Community Edition

Getting presets out of plugins



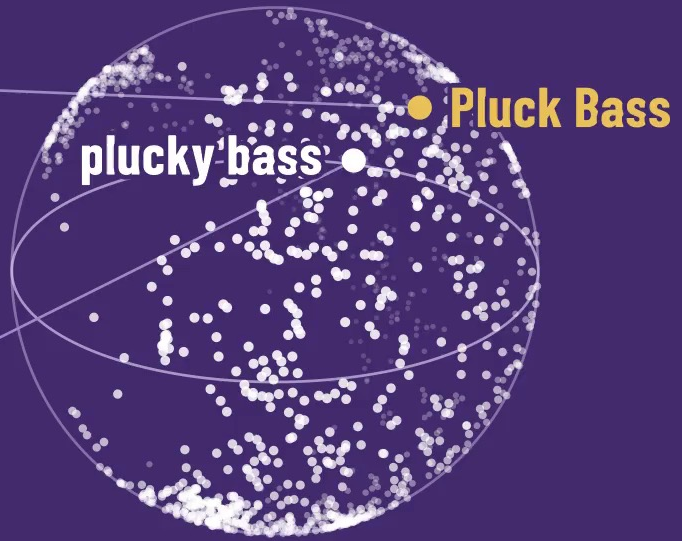
a preset's audio

audio encoder

"plucky bass"

a description

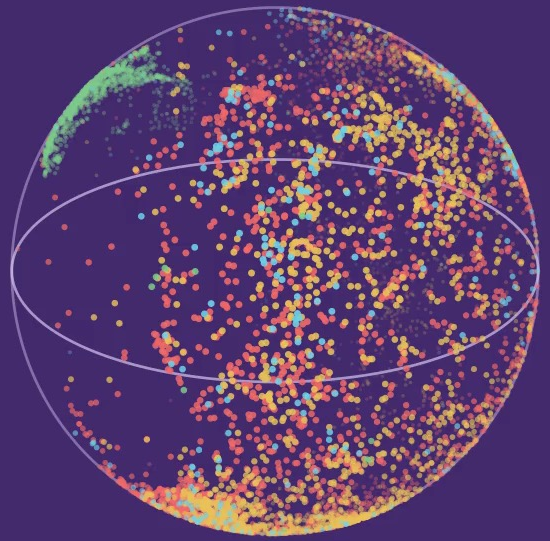
text encoder



Animations: Manim Community Edition

Getting presets out of plugins

Surge XT	2,998
Vital	2,228
Dexed	1,056
Zebralette3	382
presets	**6,664**



Animations: Manim Community Edition

One preset, two renders

Animations: Manim Community Edition

One preset, two renders

Clip 1 held C3, then short C3, E3, G3



Animations: Manim Community Edition

One preset, two renders

Clip 2 each note as a major triad



Animations: Manim Community Edition

One preset, two renders

Pluck Bass Vital

Clip 1 held C3, then short C3, E3, G3



Animations: Manim Community Edition

One preset, two renders

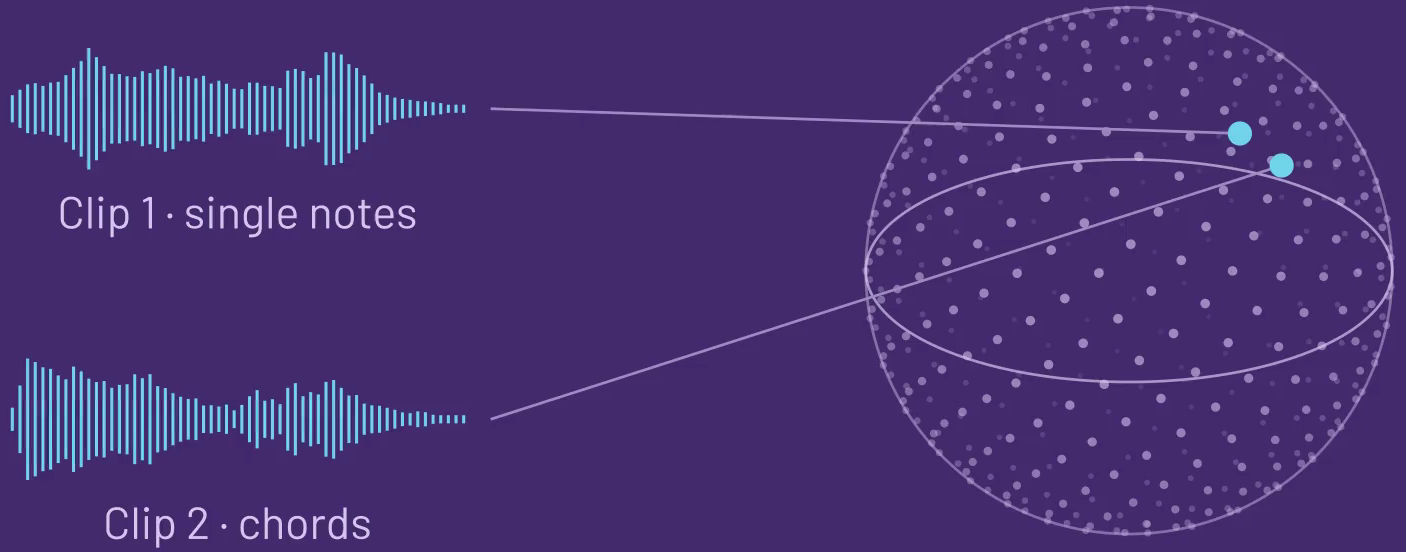
Warm Blanket Zebralette3 **Clip 2** each note as a major triad



Animations: Manim Community Edition

Search by sound or by words

Warm Blanket Zebralette3



1 preset → 2 points

The CLAP checkpoint was trained on LAION-Audio-630K (633,526 audio-text pairs), AudioSet and music. Source: LAION-CLAP README.

Animations: Manim Community Edition

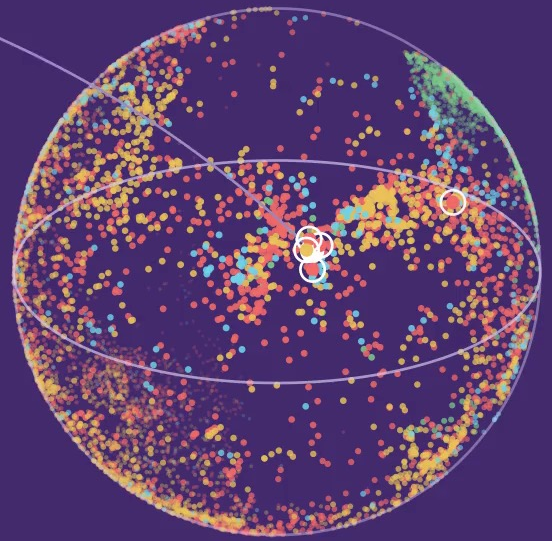
Search by sound or by words



Asymptote

a Surge XT recording

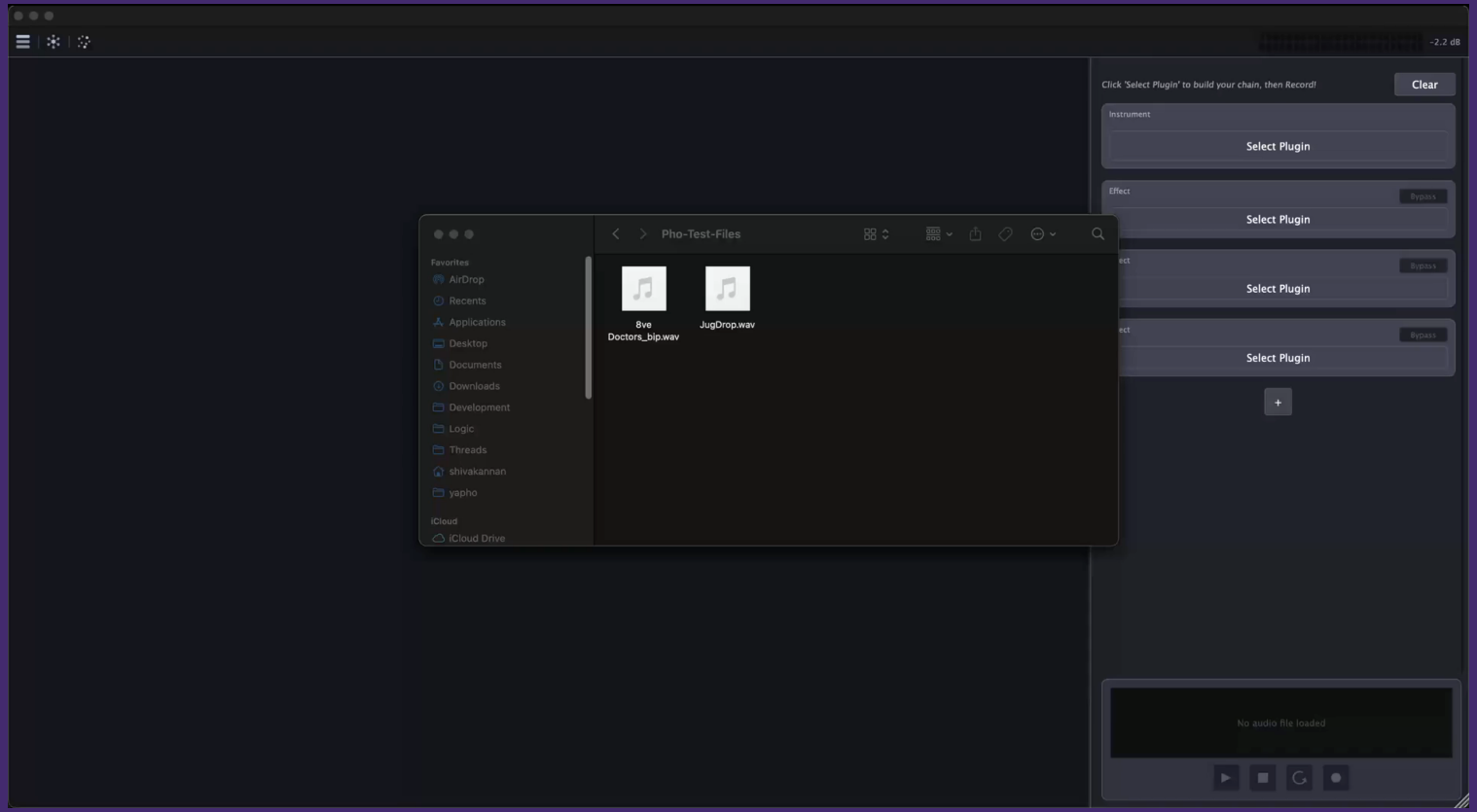
nearest presets		cos θ
1	Asymptote (itself)	**1.00**
2	Key Station	**0.94**
3	BA Trance Bass 07	**0.92**
4	Natural Bass	**0.92**
5	Plain	**0.91**



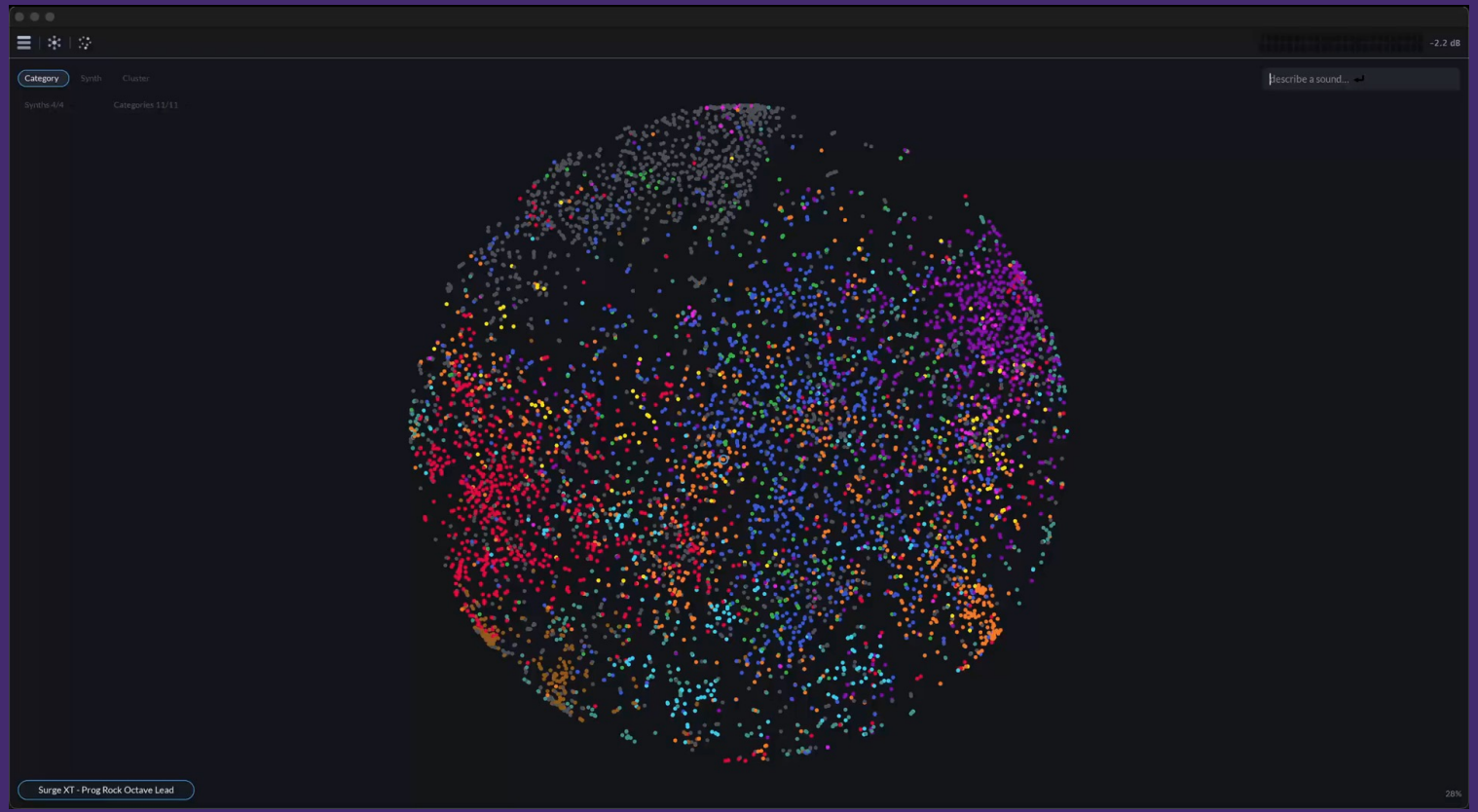
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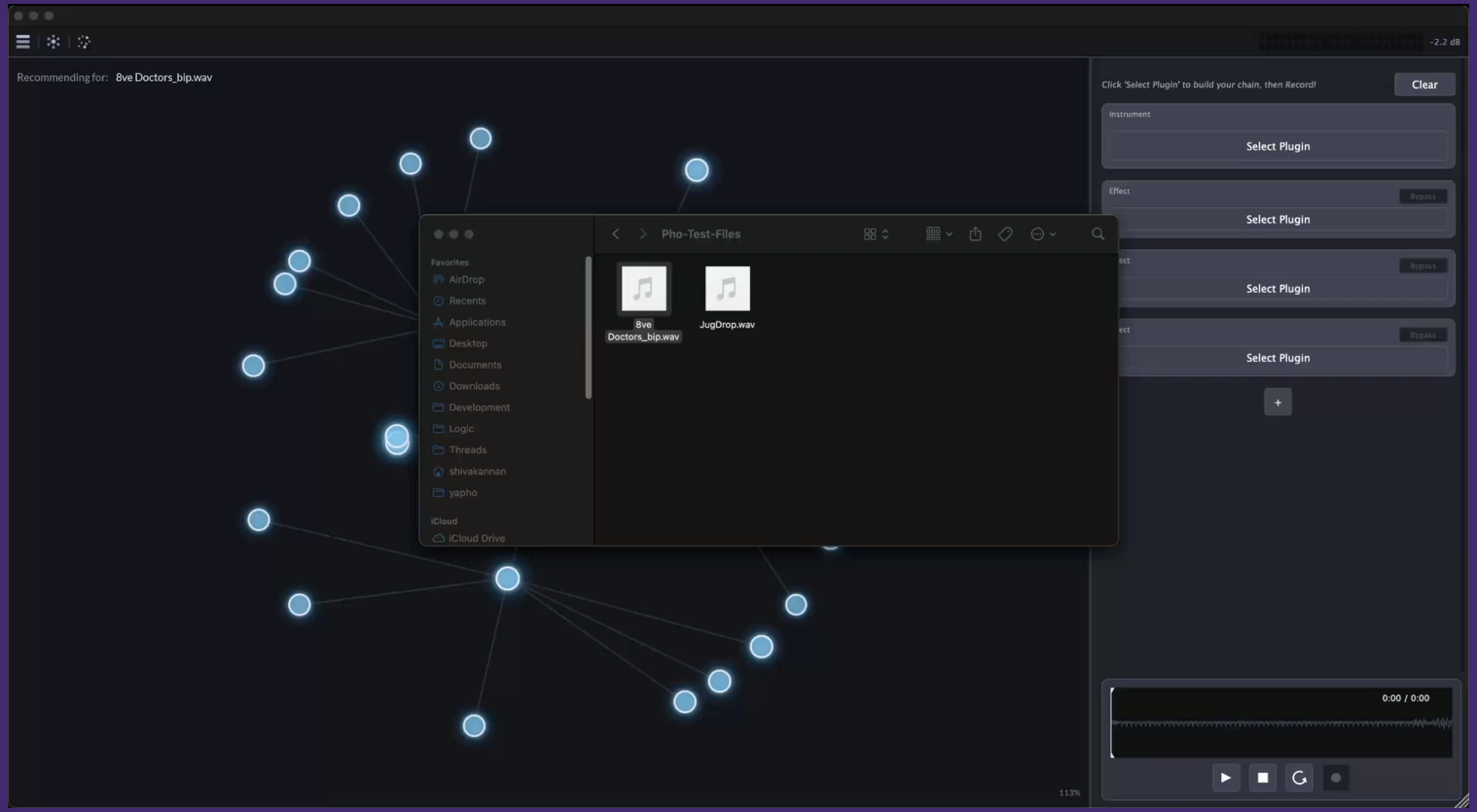
Demo: Sound Match and the Explorer



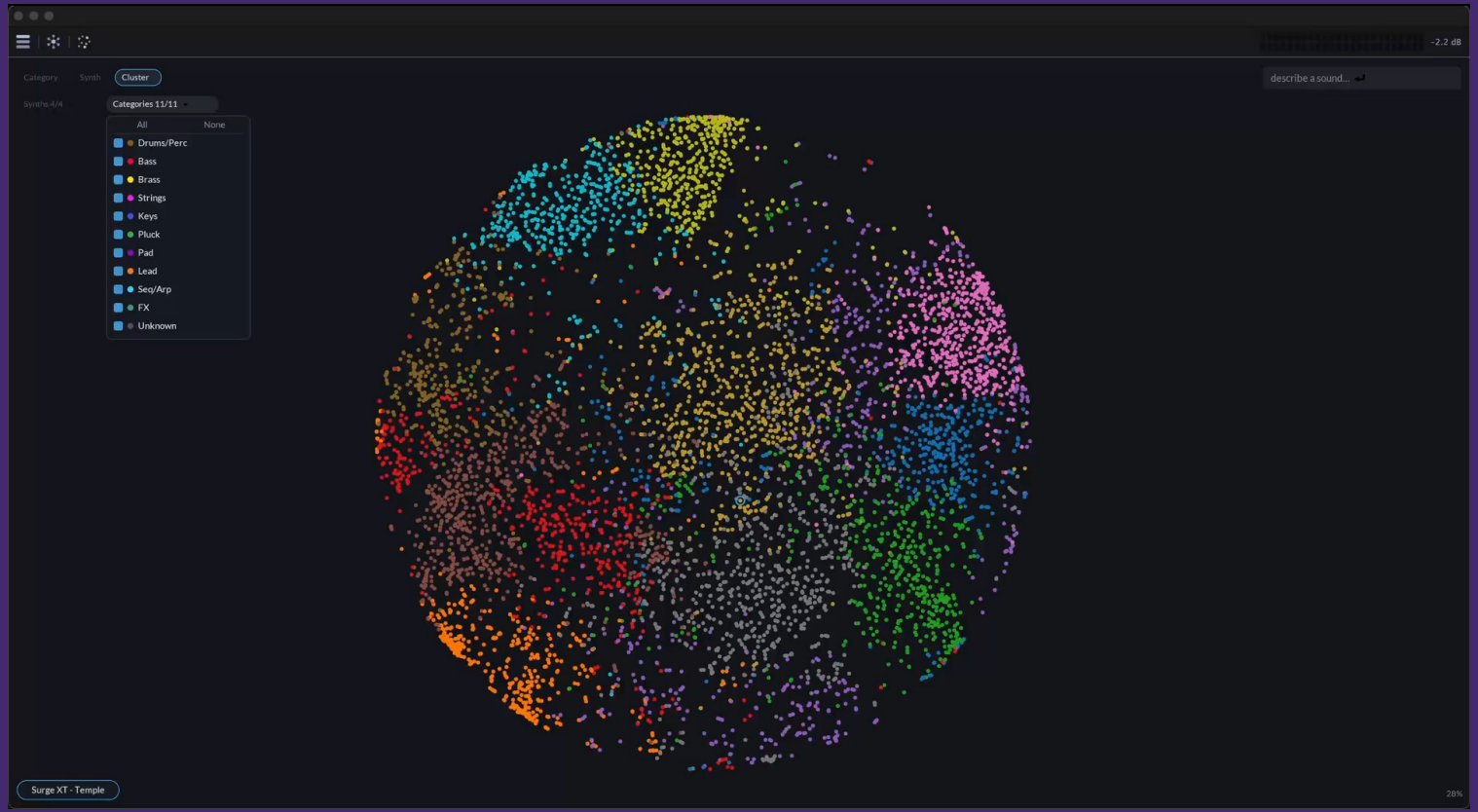
Demo: Explorer



The space between presets



The space between presets



What developers and the community can do

PLUGIN DEVELOPERS

Preset discovery

The CLAP plugin format lets a host list a synth's presets, with no file parsing.

```
// preset-discovery.h
begin_preset(name, load_key)
add_creator(creator)
set_description(description)
add_feature(feature)
```

Code: CLAP SDK, include/clap/factory/preset-discovery.h (free-audio/clap, MIT), abridged

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Mark what can morph

Cutoff	**blend**
Filter type	**switch**
Wavetable	**out of reach**

No plugin format flags this today.

Code: CLAP SDK, include/clap/factory/preset-discovery.h (free-audio/clap, MIT), abridged

What developers and the community can do

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Filter type	**switch**
Wavetable	**out of reach**

No plugin format flags this today.

MODELS AND DATA

A model that knows synth timbre

LAION-CLAP learned from captions like

"rain on a window"
"a rock song"

Needed: synth sounds described by timbre.

Code: CLAP SDK, include/clap/factory/preset-discovery.h (free-audio/clap, MIT), abridged

Further reading, and the free Pho beta

- LAION-CLAP
- Timbre semantics in CLAP
- TimbreCLIP
- SynthScribe
- CTAG
- Neural Proxies
- FM-SynAPSE
- Synth-JEPA



shiva-kannan.github.io/pho-beta

Thanks to: Rishikesh Daoo, Derek Jones, Eric Yeargan, Anirudh Venkatesh, Karran Pandey, Dinesh P